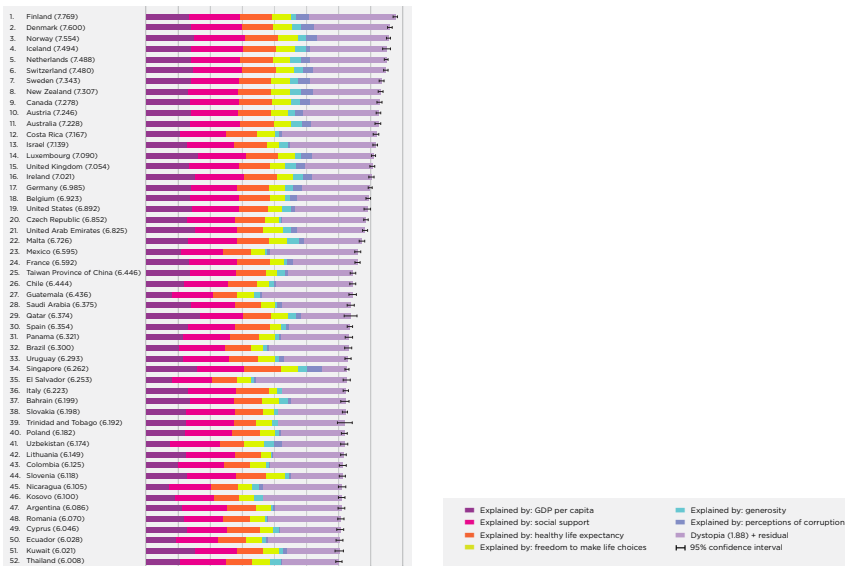


Utility

JEB104 Microeconomics I
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Can we buy happiness? Ranking of Happiness 2016–2018



Utility Function

Utility – way to describe what individual prefers

Utility function

- assigns a number to every possible consumption bundle
- higher numbers for more-preferred bundles

Only **rational** and **continuous** preference relation can be represented by a continuous utility function.

A utility function $u(\cdot)$ represents preference relation $\succeq$ if and only if:

- $x \succ y \Leftrightarrow u(x) > u(y)$
- $y \succ x \Leftrightarrow u(y) > u(x)$
- $x \sim y \Leftrightarrow u(x) = u(y)$

Utility Function: Cardinal Utility

Cardinal utility concept

- utility is cardinally or quantitatively measured in units of utility “utils”
- u : consumption bundle $\rightarrow \mathbb{R}$
- utility difference between 2 bundles matters
- Example: $u(1 \text{ banana}) = 10$, $u(1 \text{ apple}) = 5 \Rightarrow$ a banana is twice as good as an apple
- Problem with interpersonal comparisons: Is 1 util for person 1 equivalent to 1 util of person 2?

To model choices of individuals:

- We need to know what they choose given a set of alternatives
- We mainly need to know how they **rank** choices
- Not necessary to know how many utils individuals derive

Utility Function: Ordinal Utility

Ordinal utility concept

- ranking or preference orderings of alternative bundles
- size of utility not important

Example: $x \succ y \succ z$

	u_1	u_2	u_3
x	3	100	20
y	2	80	2
z	1	50	-5

Not unique utility function for given preference relation

Monotonic Transformation

Monotonic transformation

Let $I \in R$, then $f : I \rightarrow R$ is a monotonic transformation if $f(\cdot)$ is a strictly increasing function on I .

*If $f(u)$ is any **monotonic transformation** of $u(x)$ representing the given preference relation $\succeq$, then $f(u(x))$ is also a utility function representing the preference relation $\succeq$, i.e. $u(x)$ and $f(u(x))$ exhibit identical preference rankings.*

What does monotonic transformation do with shape of indifference curves?
Nothing

Monotonic Transformation

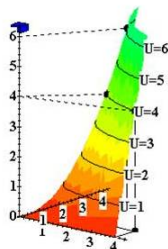
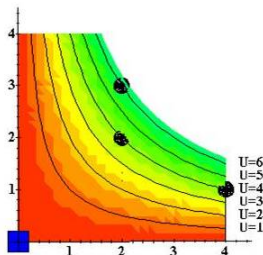
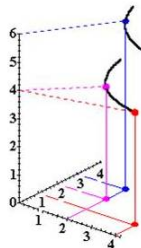
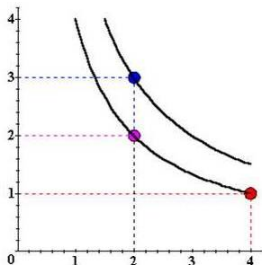
Examples of monotonic transformation:

- 1 $f(u) = u + 1$
- 2 $f(u) = 2u$
- 3 $f(u) = \exp(u)$
- 4 $f(u) = \ln(u)$ if $u > 0$
- 5 $f(u) = u^2$ if $u \geq 0$
- 6 $f(u) = u^3$ if $u \geq 0$
- 7 $f(u) = -\frac{1}{u}$ if $u > 0$
- 8 $f(u) = |u|$ if $u \geq 0$

Examples of non-monotonic transformation:

- 1 $f(u) = 10 - u$
- 2 $f(u) = u(0.5 - u)(1 - u)$
- 3 $f(u) = \sin(u)$

Constructing utility function from indifference curves



Collection of indifference curves = indifference map = utility function

Properties of Utility Function I

Continuous

- follows from continuity of preference relation
- if $\succeq$ is continuous $\Rightarrow$ *some* continuous utility function representing $\succeq$ exists
- not all utility functions representing $\succeq$ are continuous!!

Smooth

- $u(x)$ of class C^∞ , i.e. $u(x)$ has derivatives of all orders

Properties of Utility Function II

Strictly quasiconcave

- For any $x, y \in X$, $x \neq y$ and $k \in (0, 1)$, we have:

$$u(kx + (1 - k)y) > \min\{u(x), u(y)\}$$

- Follows from strict convexity of preferences
- Alternatively: for any $x, y \in X$, $x \sim y$, $x \neq y$, and $k \in (0, 1)$ we have:

$$u(z) = u(kx + (1 - k)y) > u(x) = u(y)$$

Notes:

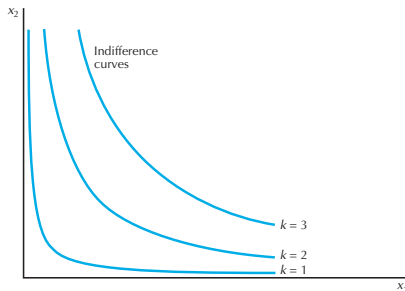
- Quasiconcavity is preserved by monotonic transformation
- We cannot say that a utility function is concave (e.g. $u(x) = x^2$)
- If it is concave, it is also quasiconcave (but not the other way around!)

To read more on quasiconcavity and quasiconcavity in notes by M.J. Osborne follow [this link](#).

Examples of Utility Functions

$$u(x, y) = x \cdot y$$

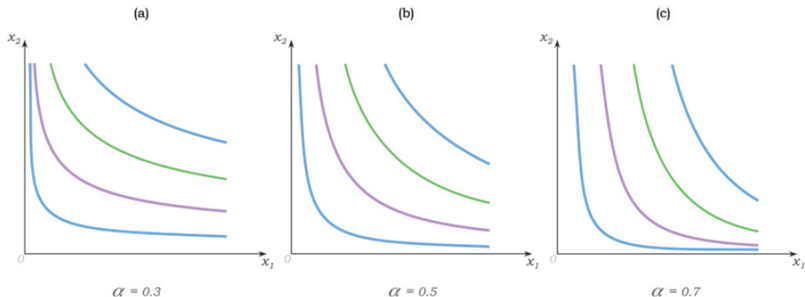
How do indifference curves look like?



And what if $u(x, y) = x^2 \cdot y^2$?

Examples of Utility Functions

Cobb-Douglas utility function: $u(x_1, x_2) = x_1^\alpha \cdot x_2^{1-\alpha}$, $\alpha \in \langle 0, 1 \rangle$



Examples of Utility Functions

Perfect substitutes

$$u(x, y) = ax + by, \quad a, b > 0$$

Perfect complements

$$u(x, y) = \min\{ax, by\}, \quad a, b > 0$$

Quasilinear preferences = linear only in one good

$$u(x, y) = f(x) + y$$

Each indifference curve is vertical displacement of each other along the axis of y .

Marginal Utility

Marginal = incremental

Marginal utility with respect to good i

- change of total utility $u(x)$ associated with a 'very small' change in the amount of x_i

$$MU_i = \frac{\partial u(x)}{\partial x_i}$$

Marginal Utility

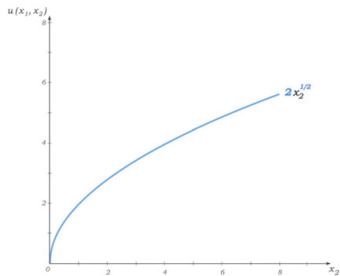
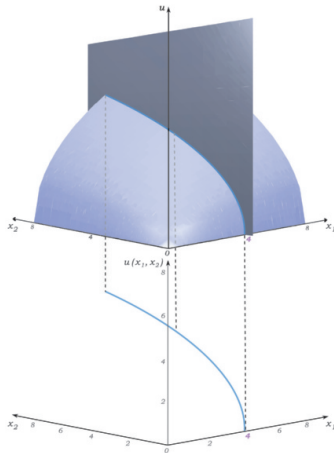
$$u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$$

(a)

$$v(x_2) = u(4, x_2) = 2x_2^{1/2}$$

$$dv/dx = x_2^{-1/2}$$

(b)



Marginal Utility

A different example: $u(x_1, x_2) = x_1^{0.5} \cdot x_2^2$

$$MU_1 = \frac{\partial u(x_1, x_2)}{\partial x_1} = 0.5 \frac{x_2^2}{x_1^{0.5}} \quad MU_2 = \frac{\partial u(x_1, x_2)}{\partial x_2} = 2x_1^{0.5} \cdot x_2$$

Potential signs of marginal utility:

- $MU_i > 0$ (most of goods)
- $MU_i < 0$ (bads)
- $MU_i = 0$ (neutral goods)

Marginal Utility and MRS

For each indifference curve: $u(x_1, x_2) = k$, a constant

We need to find (small) changes dx_1, dx_2 that keep utility constant

Total differential:

$$du = \frac{\partial u(x_1, x_2)}{\partial x_1} dx_1 + \frac{\partial u(x_1, x_2)}{\partial x_2} dx_2 = 0$$

$$\frac{dx_2}{dx_1} = - \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}} = - \frac{MU_1}{MU_2}$$

$\Downarrow$

$$MRS = - \frac{MU_1}{MU_2}$$

Diminishing MRS is a consequence of diminishing MU!

MRS and Specific Examples of Utility Functions

Perfect substitutes: $u(x_1, x_2) = ax_1 + bx_2$, $a, b > 0$

$$MRS = -\frac{a}{b}$$

Quasilinear preferences: $u(x_1, x_2) = f(x_1) + x_2$

$$MRS = -f'(x_1)$$

How does MRS change when a monotonic transformation to a utility function is applied? It does not.